Data Analyst Internship Project

Title:

Employee Engagement, Satisfaction, and Burnout Diagnostic Analysis at Palo Alto Networks

A Project Report Submitted for the Unified Mentor Internship Program

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TABLE OF CONTENTS

1. ABSTRACT

Employee engagement and well-being are critical factors that influence organizational productivity, employee retention, and overall business performance. In high-performance environments such as Palo Alto Networks, employees often face demanding workloads, frequent deadlines, and continuous pressure to deliver results. While such conditions drive innovation and growth, they also increase the risk of employee burnout, dissatisfaction, and disengagement. Traditional HR practices often address these issues only after employees decide to leave, resulting in reactive decision-making and higher organizational costs.

This project aims to develop a data-driven approach for analyzing employee engagement, satisfaction, and burnout risk using employee-related data. The primary objective is to identify early warning signals of disengagement and burnout before they lead to attrition. By leveraging key variables such as job satisfaction, work-life balance, overtime, and career progression, the project provides a comprehensive understanding of employee experience.

A key contribution of this study is the development of a **composite Engagement Index**, which combines multiple satisfaction metrics, including job involvement, job satisfaction, environment satisfaction, and relationship satisfaction. This index provides a holistic measure of employee engagement across the organization. In addition, a **Burnout Risk Score** is constructed based on overtime status and work-life balance to identify employees who are at high risk of burnout.

Exploratory Data Analysis (EDA) is conducted to uncover patterns and relationships between various factors influencing employee well-being. The analysis highlights the significant impact of overtime and poor work-life balance on employee engagement and burnout. Visualization techniques are used to present insights in a clear and interpretable manner.

Furthermore, an interactive Streamlit dashboard is developed to provide real-time visualization of engagement levels, burnout risk, and key trends across departments and job roles. This dashboard enables HR managers to monitor employee well-being, identify high-risk groups, and implement targeted interventions.

Overall, this project demonstrates how data analytics can be used to transform HR practices by shifting from reactive attrition analysis to proactive employee experience management. By identifying engagement gaps and burnout risks early, organizations can improve employee satisfaction, enhance productivity, and achieve sustainable growth.

2. INTRODUCTION

Employee engagement has emerged as one of the most critical factors influencing organizational success in the modern workplace. In highly competitive and fast-paced industries such as cybersecurity, organizations like Palo Alto Networks rely heavily on their workforce to maintain innovation, efficiency, and operational excellence. However, the increasing demands placed on employees often lead to stress, fatigue, and eventually burnout, which negatively impacts both individual performance and organizational outcomes.

Burnout is not an instantaneous event but rather a gradual process that develops over time due to continuous exposure to high workload, lack of work-life balance, and insufficient support systems. Employees experiencing burnout often exhibit symptoms such as reduced productivity, disengagement from work, and decreased job satisfaction. If left unaddressed, these issues can lead to higher attrition rates, increased recruitment costs, and loss of valuable talent.

Traditional human resource management approaches typically focus on analyzing attrition after it has already occurred. This reactive approach limits the ability of organizations to prevent employee turnover effectively. Therefore, there is a growing need for proactive systems that can identify early warning signs of disengagement and burnout before employees decide to leave the organization.

This project aims to address this challenge by leveraging data analytics techniques to analyze employee engagement, satisfaction, and burnout risk. By utilizing employee data such as job satisfaction, work-life balance, overtime status, and career progression metrics, the project seeks to uncover patterns and relationships that influence employee well-being.

A key contribution of this project is the development of an Engagement Index, which combines multiple satisfaction-related metrics into a single comprehensive score. This index provides a holistic view of employee engagement, enabling organizations to assess overall workforce health. Additionally, a Burnout Risk Score is introduced to identify employees who are at high risk of burnout based on factors such as excessive overtime and poor work-life balance.

Furthermore, the project includes the development of an interactive Streamlit dashboard that allows HR professionals to visualize insights and monitor employee engagement in real time. This dashboard serves as a decision-support tool, enabling organizations to implement targeted interventions and improve employee experience.

3. PROBLEM STATEMENT

In high-performance organizations such as Palo Alto Networks, employees are expected to operate in a fast-paced, demanding environment that requires continuous productivity, innovation, and adaptability. While such environments contribute to organizational growth and competitiveness, they also increase the likelihood of employee stress, dissatisfaction, and burnout. Over time, these factors can significantly impact employee engagement levels and lead to higher attrition rates.

One of the major challenges faced by organizations is the lack of a **unified system to measure employee engagement and well-being**. Although various individual metrics such as job satisfaction, work-life balance, and performance ratings are available, they are often analyzed in isolation. This fragmented approach makes it difficult to gain a holistic understanding of employee experience and identify underlying issues affecting engagement.

Another critical issue is the **absence of early detection mechanisms for burnout**. Burnout is a gradual process that develops due to prolonged stress, excessive workload, and poor work-life balance. However, most organizations only recognize burnout after it has already impacted employee performance or resulted in attrition. This reactive approach limits the ability of HR teams to take timely corrective actions.

Additionally, there is limited visibility into **which employee groups are more vulnerable to burnout and disengagement**. Factors such as overtime, frequent business travel, long commuting distances, and lack of career progression can significantly influence employee well-being. However, without proper analysis, these factors remain unexamined, preventing organizations from identifying high-risk segments.

Another challenge is the lack of understanding of **how different variables interact with each other**. For instance, the relationship between overtime and work-life balance, or between job satisfaction and engagement, is often not clearly analyzed. This makes it difficult to determine the root causes of employee dissatisfaction and design effective intervention strategies.

Due to these limitations, HR decisions are often based on intuition rather than data-driven insights. This leads to **late-stage interventions**, where actions are taken only after employees have already decided to leave or become disengaged. Such reactive strategies result in increased recruitment costs, loss of skilled employees, and reduced organizational efficiency.

4. OBJECTIVES

The primary objective of this project is to develop a data-driven analytical framework to evaluate employee engagement, satisfaction, and burnout risk within an organization. By leveraging employee-related data, the project aims to provide meaningful insights that support proactive human resource management and improve overall employee well-being.

One of the key objectives is to measure employee engagement using a unified metric. Instead of analyzing individual satisfaction variables separately, the project introduces a composite Engagement Index that combines multiple factors such as job involvement, job satisfaction, environment satisfaction, and relationship satisfaction. This unified approach enables a more comprehensive understanding of employee engagement across the organization.

Another important objective is to identify employees who are at risk of burnout. Burnout is often caused by excessive workload and poor work-life balance. By analyzing overtime patterns and work-life balance scores, the project aims to classify employees into different burnout risk categories, such as low, medium, and high risk. This helps HR teams take timely actions to support employees and prevent long-term negative outcomes.

The project also aims to analyze the relationship between workload and employee well-being. Factors such as overtime, business travel, and commuting distance can significantly influence stress levels and engagement. By examining these variables, the project seeks to understand how workload affects employee satisfaction and productivity.

Another objective is to study engagement trends across different employee segments, such as job roles, departments, and experience levels. This analysis helps in identifying specific groups that may be experiencing lower engagement or higher burnout risk. Understanding these patterns enables organizations to design targeted interventions and improve employee experience.

In addition, the project aims to explore the connection between employee engagement and attrition. By comparing engagement levels of employees who stayed with those who left, the project provides insights into how disengagement may lead to employee turnover. This helps organizations address potential issues before they result in attrition.

5. EXISTING SYSTEM

In most organizations, employee engagement and well-being are monitored using traditional human resource management practices that rely on periodic surveys, performance reviews, and basic reporting metrics. While these methods provide some level of insight into employee satisfaction, they are often limited in scope and fail to capture the complete picture of employee experience.

The existing system primarily depends on annual or quarterly employee surveys to assess satisfaction levels. These surveys typically include questions related to job satisfaction, work environment, and management support. However, the results are often analyzed at a high level and do not provide real-time insights into employee engagement. Additionally, survey responses may not always reflect the true sentiments of employees due to bias or hesitation in sharing honest feedback.

Another limitation of the existing system is the lack of integration between different data sources. Employee-related data, such as performance metrics, attendance records, overtime, and satisfaction scores, are often stored in separate systems. This fragmented approach makes it difficult to analyze relationships between different factors affecting employee engagement and burnout.

The current approach also lacks a standardized metric to measure engagement. While individual indicators such as job satisfaction or performance ratings are available, there is no unified index that combines multiple factors into a single, interpretable score. This makes it challenging for HR teams to assess overall engagement levels and compare them across departments or employee groups.

A significant drawback of the existing system is its reactive nature. Organizations typically identify issues such as burnout or disengagement only after they have already impacted employee performance or resulted in attrition. For example, high attrition rates may indicate underlying problems, but by the time these issues are recognized, it is often too late to retain affected employees.

Furthermore, there is limited capability to identify high-risk employee groups. Factors such as overtime, poor work-life balance, long commuting distances, and lack of career growth opportunities can contribute to burnout. However, without advanced analytical tools, these factors are not systematically analyzed, and high-risk employees remain unidentified.

6. PROPOSED SYSTEM

To overcome the limitations of the existing system, this project proposes a data-driven analytical framework for evaluating employee engagement, satisfaction, and burnout risk. The proposed system integrates multiple employee-related factors into a unified platform, enabling organizations to gain a comprehensive understanding of employee well-being and take proactive measures to improve it.

The core component of the proposed system is the development of a composite Engagement Index, which combines multiple satisfaction-related attributes such as job involvement, job satisfaction, environment satisfaction, and relationship satisfaction. Unlike the existing system, where these factors are analyzed individually, the Engagement Index provides a single, standardized metric that represents overall employee engagement. This makes it easier for HR managers to compare engagement levels across departments, job roles, and experience levels.

Another key feature of the proposed system is the introduction of a Burnout Risk Classification model. This model identifies employees who are at risk of burnout based on their overtime status and work-life balance scores. Employees who frequently work overtime and report low work-life balance are categorized as high-risk, while others are classified as medium or low risk. This classification enables early detection of burnout and allows HR teams to take timely interventions before the situation worsens.

The proposed system also incorporates a Workload Stress Indicator, which evaluates employee stress levels based on factors such as overtime and business travel frequency. This indicator provides additional insights into how workload-related factors influence employee well-being and helps in identifying employees who may be experiencing excessive stress.

In addition to KPI development, the system performs Exploratory Data Analysis (EDA) to uncover patterns and relationships between various factors affecting employee engagement. For example, the system analyzes how overtime impacts engagement levels, how work-life balance influences satisfaction, and how engagement varies across job roles and career stages. These insights provide valuable information for improving HR policies and practices.

A major advantage of the proposed system is the development of an interactive Streamlit dashboard, which serves as a user-friendly interface for visualizing insights. The dashboard includes multiple modules such as Engagement Overview, Burnout Risk Analysis, Role and Career Stage Analysis, and Manager Action Panel. Users can apply filters based on department, job role, and experience level to explore specific segments of the workforce.

The system also supports real-time analysis, allowing HR managers to monitor employee engagement and burnout risk continuously. This enables organizations to shift from reactive decision-making to proactive workforce management, where issues are identified and addressed before they lead to negative outcomes such as attrition.

Furthermore, the proposed system provides actionable recommendations based on the analysis. For example, employees identified as high-risk can be provided with workload adjustments, flexible working options, or additional support. Similarly, departments with low engagement scores can be targeted for improvement initiatives.

Overall, the proposed system offers a comprehensive, scalable, and data-driven solution for managing employee engagement and well-being. By integrating multiple data sources, developing meaningful KPIs, and providing interactive visualizations, the system empowers organizations to make informed decisions and create a healthier, more productive work environment.

7. DATASET DESCRIPTION

The dataset used in this project contains comprehensive information about employees, including demographic details, job-related attributes, satisfaction metrics, and performance indicators. This dataset plays a crucial role in analyzing employee engagement, satisfaction, and burnout risk, as it provides multiple dimensions of employee experience within the organization.

The dataset consists of both **numerical and categorical variables**, allowing for a detailed analysis of employee behavior and workplace dynamics. Each record in the dataset represents an individual employee, capturing various aspects of their professional and personal work environment.

◆ Demographic and Personal Attributes

The dataset includes basic demographic information such as **Age** and **Gender**, which help in understanding how engagement and burnout patterns vary across different employee groups. Additionally, **Marital Status** is included to analyze whether personal life factors influence work-life balance and overall satisfaction.

◆ Job and Organizational Attributes

Several fields describe the employee’s role within the organization. These include **Department**, **Job Role**, and **Job Level**, which indicate the employee’s position and responsibilities. These attributes are important for analyzing how engagement levels differ across departments and hierarchical levels.

The dataset also includes **Business Travel**, which indicates how frequently employees are required to travel for work. Frequent travel can contribute to increased stress and reduced work-life balance, making it an important factor in burnout analysis.

◆ Compensation and Financial Attributes

The dataset contains multiple fields related to employee compensation, such as **Daily Rate**, **Hourly Rate**, **Monthly Income**, and **Monthly Rate**. These variables help in understanding the relationship between financial rewards and employee satisfaction. Additionally, **Percent Salary Hike** and **Stock Option Level** provide insights into employee growth and financial incentives.

◆ Work Experience and Career Growth

Employee experience and career progression are captured through variables such as **Total Working Years**, **Years at Company**, **Years in Current Role**, **Years Since Last Promotion**, and **Years with Current Manager**. These attributes help in analyzing how career stagnation or growth opportunities impact employee engagement and satisfaction.

For instance, employees who have not received promotions for a long time may experience reduced motivation and engagement. Similarly, employees with longer tenure in the same role may feel stagnation, leading to dissatisfaction.

◆ Satisfaction and Engagement Metrics

A key component of the dataset is the inclusion of multiple satisfaction-related variables, such as:

- **Job Satisfaction**
- **Environment Satisfaction**
- **Relationship Satisfaction**
- **Job Involvement**

These variables are typically measured on a scale of 1 to 4, where higher values indicate greater satisfaction. These metrics form the foundation for calculating the Engagement Index and provide insights into different dimensions of employee experience.

◆ Work-Life Balance and Workload Indicators

The dataset includes **Work-Life Balance**, which is a critical factor in determining employee well-being. Employees with low work-life balance are more likely to experience stress and burnout.

Another important variable is **OverTime**, which indicates whether an employee works beyond regular working hours. Overtime is a key contributor to burnout and is used in the calculation of the Burnout Risk Score.

Additionally, **Distance from Home** provides insights into commuting stress, which can also impact employee satisfaction and work-life balance.

Performance and Training Metrics

The dataset includes **Performance Rating**, which reflects the employee's performance level, and **Training Times Last Year**, which indicates the number of training programs attended. These variables help in understanding how performance and skill development influence engagement and career growth.

Attrition Indicator

The **Attrition** variable indicates whether an employee has left the organization. This variable is useful for analyzing the relationship between engagement levels and employee turnover. By comparing engagement scores of employees who stayed versus those who left, valuable insights can be derived regarding factors leading to attrition.

Importance of the Dataset

This dataset provides a holistic view of employee experience by combining demographic, professional, and satisfaction-related information. It enables a multi-dimensional analysis of employee engagement and burnout risk, making it possible to identify patterns and trends that are not immediately visible.

By leveraging this dataset, the project is able to:

- Analyze relationships between workload, satisfaction, and engagement
- Identify high-risk employee groups
- Understand the impact of career growth and compensation
- Support data-driven HR decision-making

8. DATA SCIENCE METHODOLOGY

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9. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is a crucial step in understanding the underlying patterns, relationships, and trends within the employee dataset. It helps in identifying key factors that influence employee engagement, satisfaction, and burnout risk. In this project, EDA is performed using a combination of statistical techniques and visualizations to derive meaningful insights from the data.

1. Distribution of Employee Satisfaction Metrics

The analysis begins by examining the distribution of satisfaction-related variables, including Job Satisfaction, Environment Satisfaction, Relationship Satisfaction, and Job Involvement. These variables are measured on a scale of 1 to 4, where higher values indicate greater satisfaction.

The distribution analysis reveals that most employees fall within the mid-range of satisfaction levels, indicating moderate engagement across the organization. However, a noticeable proportion of employees report low satisfaction scores, which could be an early indicator of disengagement.

This observation highlights the importance of focusing on employees with lower satisfaction scores, as they are more likely to experience reduced motivation and productivity.

2. Engagement Index Analysis

The Engagement Index, which is a composite score derived from multiple satisfaction metrics, is analyzed to understand overall engagement levels within the organization.

The distribution of the Engagement Index shows variability across employees, indicating that engagement is not uniform. While some employees exhibit high engagement levels, others fall into lower ranges, suggesting potential issues related to job satisfaction, work environment, or relationships within the organization.

Further analysis reveals that employees with higher engagement scores tend to have better job satisfaction and work-life balance, reinforcing the importance of these factors in maintaining employee well-being.

3. Overtime vs Employee Engagement

One of the most significant analyses performed in this project is the relationship between overtime and employee engagement.

The results indicate that employees who work overtime generally have lower engagement scores compared to those who do not. This suggests that excessive workload can lead to fatigue, stress, and reduced motivation.

In addition, overtime is found to be a major contributor to burnout risk. Employees who consistently work beyond regular hours are more likely to experience dissatisfaction and disengagement.

4. Work-Life Balance vs Satisfaction

Work-life balance is a critical factor influencing employee well-being. The analysis shows a strong positive relationship between work-life balance and satisfaction metrics.

Employees with higher work-life balance ratings tend to report higher job satisfaction, better relationships, and greater overall engagement. On the other hand, employees with poor work-life balance are more likely to experience stress and burnout.

This finding emphasizes the need for organizations to promote policies that support a healthy balance between professional and personal life.

5. Burnout Risk Distribution

The Burnout Risk Score is analyzed to identify the proportion of employees in different risk categories (Low, Medium, High).

The analysis reveals that a significant number of employees fall into the medium and high-risk categories, indicating potential areas of concern. High-risk employees are primarily those who work overtime and have low work-life balance.

This insight highlights the importance of monitoring burnout risk and implementing preventive measures to support affected employees.

6. Job Role vs Engagement Analysis

Engagement levels are analyzed across different job roles to identify variations in employee experience.

The results show that certain roles, particularly those with higher responsibilities and workload, tend to have lower engagement scores. This may be due to increased pressure, longer working hours, and higher expectations associated with these roles.

7. Experience and Tenure Analysis

Employee engagement is also analyzed based on experience and tenure within the organization.

The analysis indicates that employees with moderate experience levels tend to have higher engagement compared to those with very low or very high tenure. Employees with longer tenure may experience stagnation, while new employees may still be adjusting to the work environment.

This suggests the importance of providing continuous learning and growth opportunities to maintain engagement across all experience levels.

8. Attrition vs Engagement (Contextual Analysis)

Although the primary focus of this project is engagement and burnout, attrition is analyzed to understand its relationship with engagement levels.

The results show that employees who left the organization generally had lower engagement scores compared to those who stayed. This indicates that disengagement is a key factor contributing to employee turnover.

This insight reinforces the importance of identifying and addressing engagement issues early to prevent attrition.

9. Key Observations from EDA

The EDA phase provides several important insights:

- Overtime has a negative impact on employee engagement
- Poor work-life balance increases burnout risk
- Job satisfaction is a strong driver of engagement
- Certain job roles are more prone to stress and disengagement
- Engagement levels vary across experience and tenure

10. Summary of EDA

Overall, Exploratory Data Analysis plays a vital role in uncovering patterns and relationships within the dataset. It provides a deeper understanding of the factors influencing employee engagement and burnout, enabling organizations to take proactive measures to improve employee well-being.

10. MACHINE LEARNING MODELS

In addition to exploratory data analysis and KPI-based evaluation, machine learning techniques are incorporated in this project to enhance analytical capabilities and provide predictive insights. While the primary focus of the project is to analyze employee engagement and burnout risk, machine learning models enable the identification of hidden patterns within the data and support proactive decision-making by predicting outcomes such as employee attrition and engagement levels.

The implementation of machine learning begins with the selection of relevant features that influence employee behavior. Variables such as job satisfaction, work-life balance, overtime status, monthly income, years at the company, job level, and environment satisfaction are considered important predictors. These features capture various dimensions of employee experience, including emotional satisfaction, workload, financial incentives, and career progression, all of which contribute to engagement and burnout.

Before training the models, the dataset undergoes preprocessing to ensure that it is suitable for analysis. Categorical variables such as job role, department, and business travel are encoded into numerical formats using appropriate encoding techniques. Numerical variables such as income and experience are normalized to maintain consistency and comparability across features. Missing values, if present, are handled carefully to avoid bias in the model. The dataset is then divided into training and testing subsets to evaluate model performance effectively.

A baseline model is developed using logistic regression, which is widely used for classification tasks such as predicting employee attrition. This model estimates the probability of an employee leaving the organization based on input features. Logistic regression is simple, interpretable, and provides a good starting point for understanding the relationship between independent variables and the target outcome. However, its limitation lies in its assumption of linear relationships, which may not fully capture the complexity of real-world data.

To address this limitation, more advanced models such as decision trees are utilized. Decision trees are capable of modeling non-linear relationships by splitting the data based on feature values. They provide clear decision rules that can be easily interpreted, making them useful for understanding how different factors contribute to employee engagement and burnout. However, decision trees can be prone to overfitting, especially when the model becomes too complex.

To improve prediction accuracy and reduce overfitting, ensemble learning techniques such as random forest are employed. Random forest combines multiple decision trees and aggregates their predictions, resulting in a more robust and accurate model. This approach enhances generalization and reduces the impact of noise in the data. Additionally, random forest models provide feature importance scores, which help in identifying the most influential factors affecting employee engagement and attrition.

Further improvement in model performance is achieved through gradient boosting techniques, which build models sequentially by correcting the errors of previous models. Gradient boosting is highly effective in capturing complex relationships within the data and often delivers superior predictive performance. However, it requires careful tuning and higher computational resources compared to simpler models.

The performance of different models is compared using evaluation metrics, and the results indicate that ensemble models generally outperform simpler models in terms of accuracy. However, interpretability remains an important consideration in HR analytics, as decision-makers need to understand the reasoning behind predictions. Therefore, a balance between accuracy and interpretability is maintained while selecting the final model.

An important outcome of the machine learning analysis is the identification of key features that significantly influence employee engagement and burnout. Factors such as work-life balance, job satisfaction, overtime, and income emerge as critical drivers. These insights align with the findings from exploratory data analysis and further validate the importance of these variables.

Overall, the integration of machine learning models in this project enhances its analytical depth by providing predictive capabilities and uncovering complex patterns within the data. While the primary emphasis remains on descriptive and diagnostic analytics, the use of machine learning enables organizations to move towards a more proactive approach in managing employee engagement and well-being.

11. MODEL EVALUATION

The evaluation of machine learning models is a critical step in assessing their effectiveness and reliability in predicting outcomes such as employee attrition and engagement levels. In this project, multiple models are evaluated to determine their ability to accurately capture patterns within the dataset and provide meaningful predictions. The evaluation process ensures that the selected model not only performs well on training data but also generalizes effectively to unseen data.

To evaluate the performance of the models, the dataset is divided into training and testing subsets. The models are trained on the training data and then tested on the unseen test data to measure their predictive capability. This approach helps in identifying issues such as overfitting, where a model performs well on training data but fails to generalize to new data.

Several evaluation metrics are used to assess model performance. One of the primary metrics is accuracy, which measures the proportion of correct predictions made by the model. While accuracy provides a general overview of model performance, it may not always be sufficient, especially in cases where the dataset is imbalanced. Therefore, additional metrics such as precision, recall, and F1-score are also considered to provide a more comprehensive evaluation.

Precision measures the proportion of correctly predicted positive instances out of all predicted positive instances, indicating the model's ability to avoid false positives. Recall, on the other hand, measures the proportion of correctly predicted positive instances out of all actual positive instances, reflecting the model's ability to identify true positives. The F1-score combines precision and recall into a single metric, providing a balanced measure of model performance.

Another important tool used in model evaluation is the confusion matrix, which provides a detailed breakdown of prediction outcomes. It shows the number of true positives, true negatives, false positives, and false negatives, allowing for a deeper understanding of model behavior. By analyzing the confusion matrix, it is possible to identify specific areas where the model may be making errors.

For probabilistic models such as logistic regression, the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) are used to evaluate performance. The ROC curve illustrates the trade-off between true positive rate and false positive rate at different threshold values, while the AUC provides a single measure of overall model performance. A higher AUC value indicates better model performance.

During the evaluation process, it is observed that simpler models such as logistic regression provide good interpretability but may have lower predictive accuracy compared to more complex models. Decision trees offer improved performance by capturing non-linear relationships but may suffer from overfitting. Ensemble models such as random forest and gradient boosting demonstrate superior performance by reducing variance and improving generalization.

However, it is important to note that in the context of HR analytics, interpretability is as important as accuracy. Decision-makers need to understand the factors influencing predictions in order to take appropriate actions. Therefore, while selecting the final model, a balance is maintained between predictive performance and interpretability.

The evaluation results indicate that ensemble models generally achieve higher accuracy and better performance metrics compared to baseline models. At the same time, feature importance analysis from these models provides valuable insights into the key drivers of employee engagement and burnout, such as work-life balance, job satisfaction, and overtime.

Overall, the model evaluation process ensures that the selected model is robust, reliable, and capable of supporting data-driven decision-making. By carefully analyzing performance metrics and understanding model behavior, the project establishes a strong foundation for predictive analytics in employee engagement and burnout analysis.

12. RESULTS AND INSIGHTS

The analysis of employee engagement, satisfaction, and burnout risk provides several meaningful insights that highlight the underlying factors influencing employee well-being within the organization. By combining exploratory data analysis, KPI-based evaluation, and machine learning techniques, the project uncovers patterns that are critical for improving human resource strategies and organizational performance.

One of the most significant findings of this study is the strong impact of **overtime on employee engagement and burnout risk**. Employees who frequently work overtime tend to exhibit lower engagement scores and higher burnout risk levels. This indicates that excessive workload not only affects employee productivity but also contributes to long-term dissatisfaction and disengagement. The relationship between overtime and burnout emphasizes the need for organizations to monitor working hours and implement policies that prevent excessive workload.

Another important insight is the role of **work-life balance in maintaining employee satisfaction and engagement**. The analysis shows a clear positive correlation between work-life balance and engagement levels. Employees who report higher work-life balance tend to have better job satisfaction, stronger relationships within the workplace, and higher overall engagement. On the other hand, employees with poor work-life balance are more likely to experience stress, fatigue, and reduced motivation. This finding highlights the importance of flexible working arrangements and supportive workplace policies.

The study also reveals that **job satisfaction is one of the strongest drivers of employee engagement**. Employees who are satisfied with their roles are more likely to be motivated, productive, and committed to the organization. Job satisfaction is influenced by several factors, including work environment, recognition, career growth opportunities, and management support. Improving these aspects can significantly enhance employee engagement levels.

In addition, the analysis identifies variations in engagement across different **job roles and organizational levels**. Employees in roles with higher responsibilities and pressure tend to have lower engagement scores, possibly due to increased workload and stress. Similarly, employees in mid-level positions may experience stagnation, leading to reduced motivation and engagement. These findings suggest that organizations should focus on providing career development opportunities and ensuring balanced workload distribution across roles.

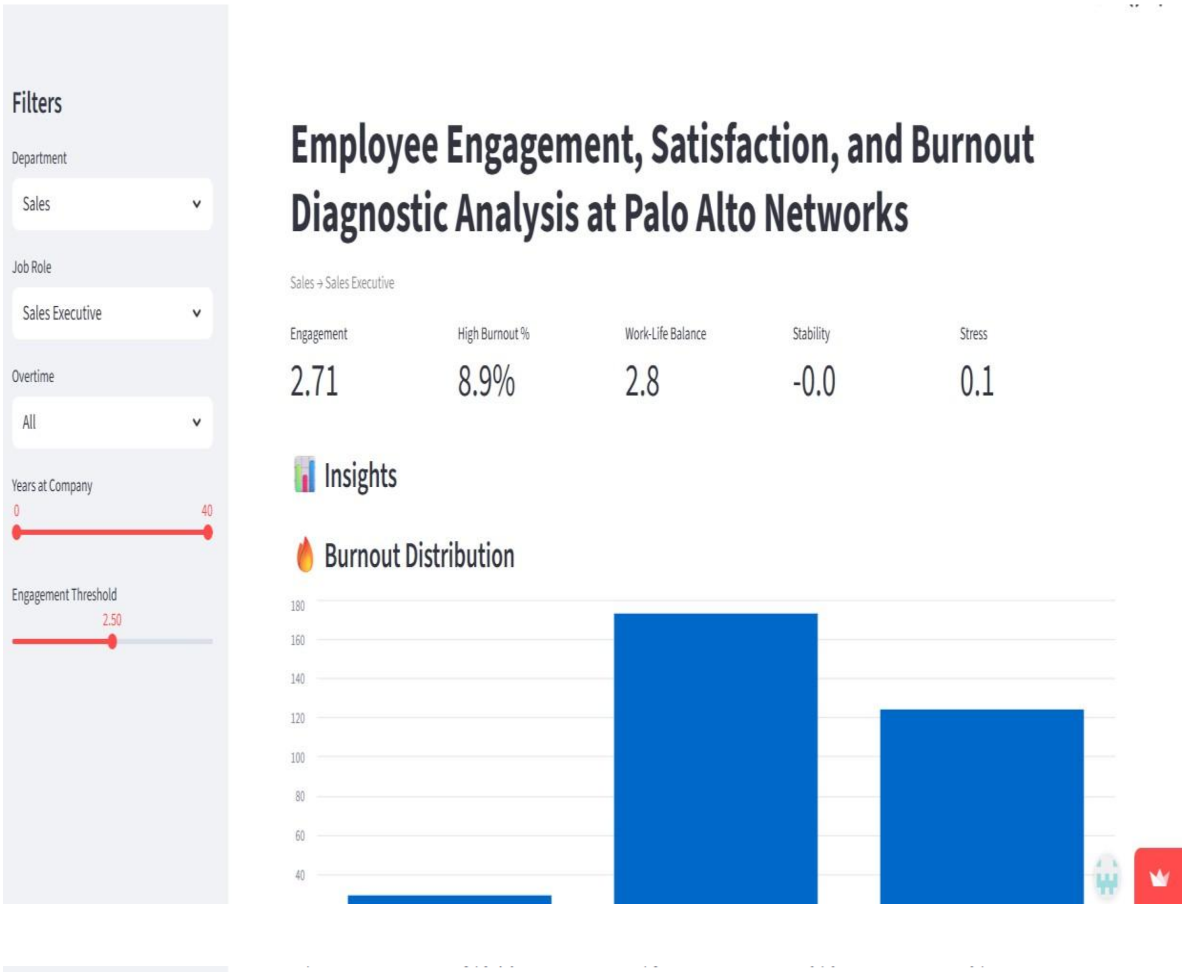
Another key insight relates to **employee tenure and experience**. The analysis indicates that employees with moderate experience levels tend to have higher engagement compared to those with very low or very high tenure. New employees may still be adapting to the work environment, while long-tenured employees may experience stagnation or lack of growth opportunities. This highlights the importance of continuous learning, skill development, and career progression programs to maintain engagement across all experience levels.

The relationship between **burnout risk and engagement levels** is also clearly observed. Employees classified as high-risk based on overtime and poor work-life balance consistently show lower engagement scores. This demonstrates that burnout and engagement are closely interconnected, and addressing burnout factors can significantly improve overall engagement.

Furthermore, the analysis of **attrition patterns** reveals that employees who left the organization generally had lower engagement scores compared to those who stayed. This confirms that disengagement is a key predictor of employee turnover. By identifying disengaged employees early, organizations can take proactive measures to retain talent and reduce attrition rates.

The Streamlit dashboard developed in this project plays a crucial role in visualizing these insights and making them accessible to decision-makers. The dashboard allows HR managers to monitor engagement levels, identify high-risk employees, and analyze trends across departments and job roles. This real-time visibility enables organizations to take timely actions and improve employee experience.

Overall, the results of this project demonstrate that employee engagement and burnout are influenced by multiple interconnected factors, including workload, work-life balance, job satisfaction, and career progression. By addressing these factors through data-driven strategies, organizations can enhance employee well-being, improve productivity, and achieve long-term success.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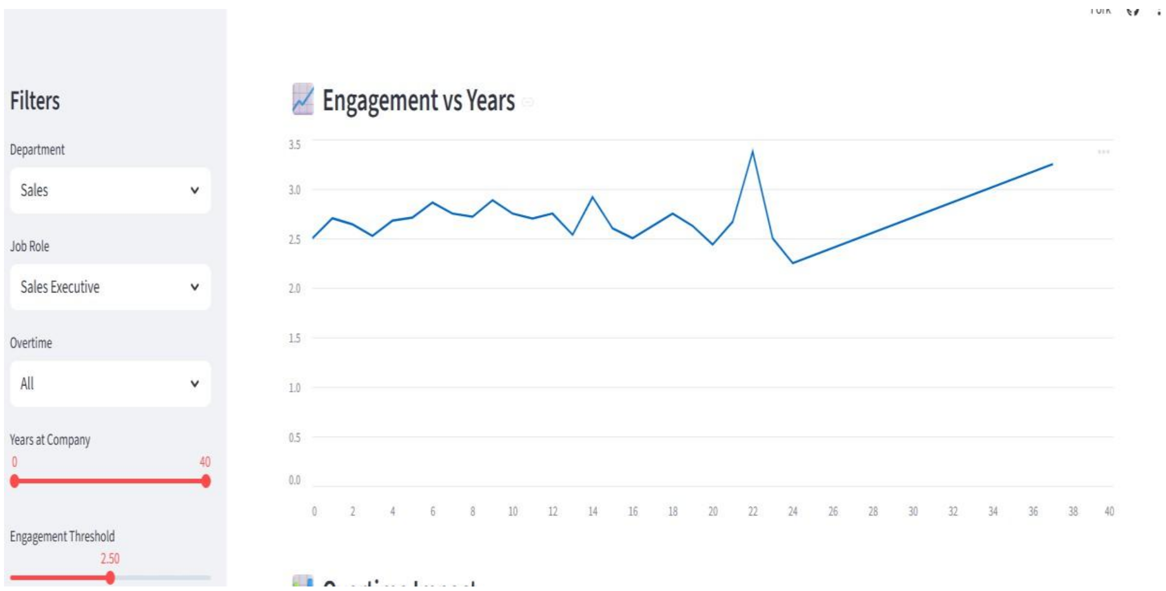
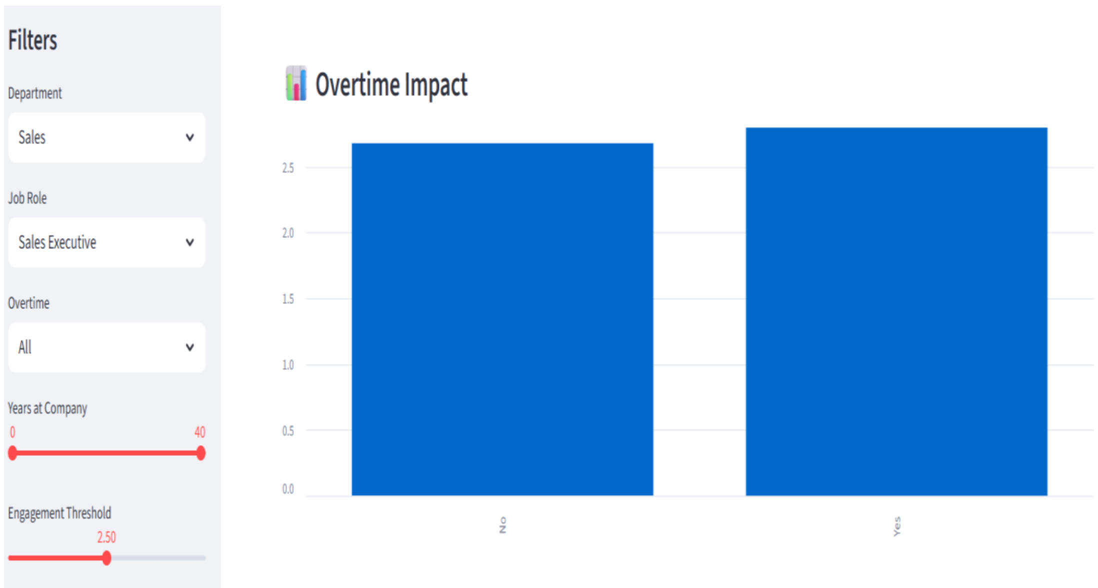
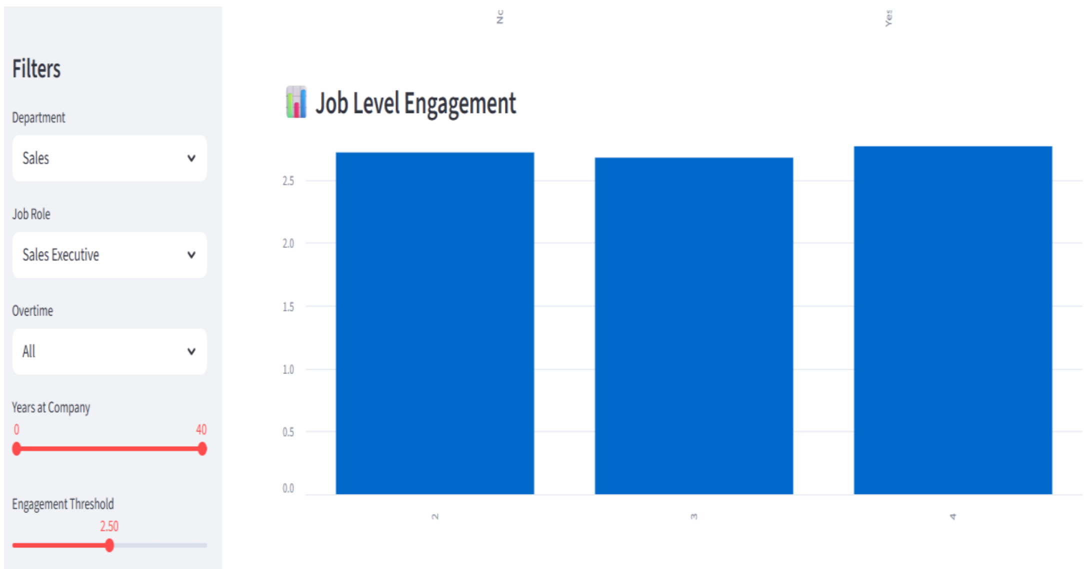
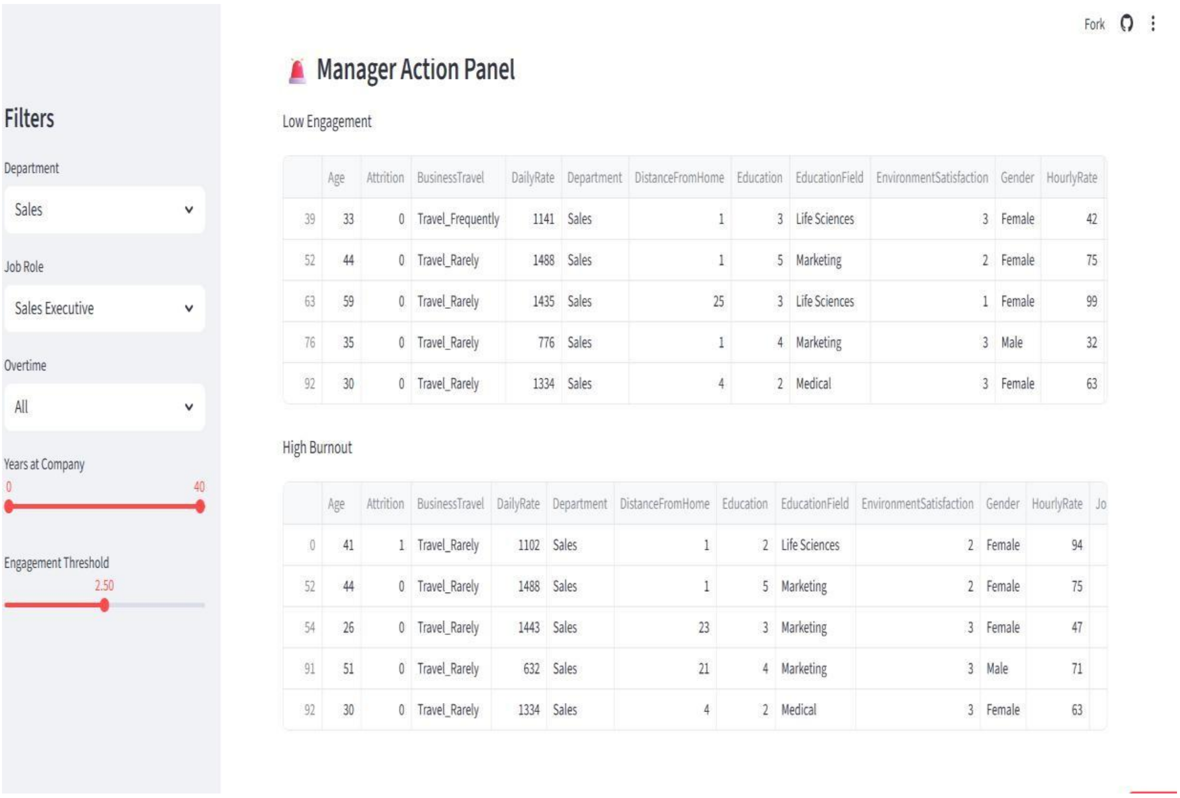
Department
Sales
Job Role
Sales Executive
Overtime
All
Years at Company
0
40
Engagement Threshold
2.50
Employee Engagement, Satisfaction, and Burnout Diagnostic Analysis at Palo Alto Networks
Sales → Sales Executive
Engagement
2.71
High Burnout %
8.9%
Work-Life Balance
2.8
Stability
-0.0
Stress
0.1
Insights
Burnout Distribution
180
160
140
120
100
80
60
40


Filters
Department
Sales
Job Role
Sales Executive
Overtime
All
Years at Company
0
40
Engagement Threshold
2.50
Insights
Burnout Distribution
180
160
140
120
100
80
60
40
20
0
High
Low
Medium


Filters
Department
Sales
Job Role
Sales Executive
Overtime
All
Years at Company
0
40
Engagement Threshold
2.50
Engagement vs Years
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40


Filters
Department
Sales
Job Role
Sales Executive
Overtime
All
Years at Company
0
40
Engagement Threshold
2.50
Overtime Impact
2.5
2.0
1.5
1.0
0.5
0.0
No
Yes

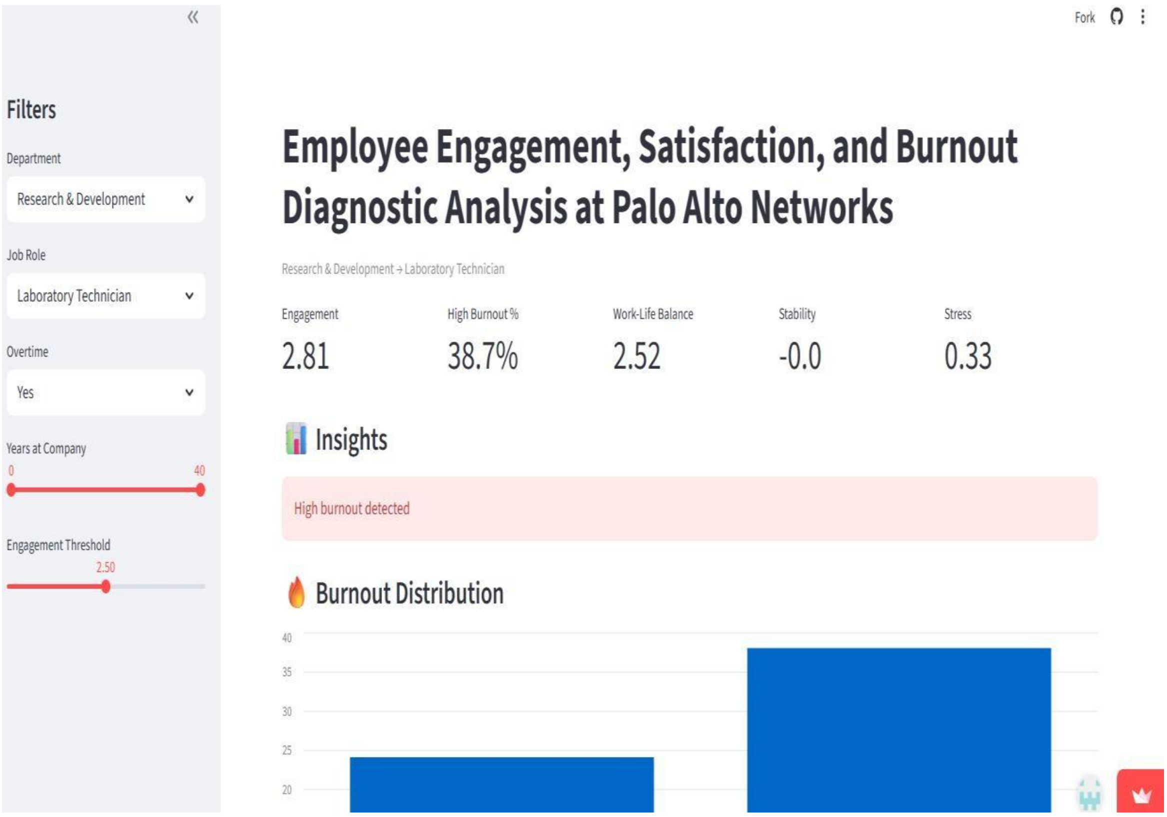

Filters
Department
Sales
Job Role
Sales Executive
Overtime
All
Years at Company
0
40
Engagement Threshold
2.50
Job Level Engagement
2.5
2.0
1.5
1.0
0.5
0.0


Fork
Filters
Department
Sales
Job Role
Sales Executive
Overtime
All
Years at Company
0
40
Engagement Threshold
2.50
Manager Action Panel
Low Engagement

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction	Gender	HourlyRate
39	33	0	Travel_Frequently	1141	Sales	1	3	Life Sciences	3	Female	42
52	44	0	Travel_Rarely	1488	Sales	1	5	Marketing	2	Female	75
63	59	0	Travel_Rarely	1435	Sales	25	3	Life Sciences	1	Female	99
76	35	0	Travel_Rarely	776	Sales	1	4	Marketing	3	Male	32
92	30	0	Travel_Rarely	1334	Sales	4	2	Medical	3	Female	63

High Burnout

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction	Gender	HourlyRate	Jo
0	41	1	Travel_Rarely	1102	Sales	1	2	Life Sciences	2	Female	94	
52	44	0	Travel_Rarely	1488	Sales	1	5	Marketing	2	Female	75	
54	26	0	Travel_Rarely	1443	Sales	23	3	Marketing	3	Female	47	
91	51	0	Travel_Rarely	632	Sales	21	4	Marketing	3	Male	71	
92	30	0	Travel_Rarely	1334	Sales	4	2	Medical	3	Female	63	


Fork
Filters
Department
Research & Development
Job Role
Laboratory Technician
Overtime
Yes
Years at Company
0
40
Engagement Threshold
2.50
Employee Engagement, Satisfaction, and Burnout Diagnostic Analysis at Palo Alto Networks
Research & Development → Laboratory Technician
Engagement
2.81
High Burnout %
38.7%
Work-Life Balance
2.52
Stability
-0.0
Stress
0.33
Insights
High burnout detected
Burnout Distribution
40
35
30
25
20

13. BUSINESS IMPACT

The implementation of this employee engagement and burnout diagnostic system has significant implications for organizational performance, human resource management, and long-term sustainability. By transforming raw employee data into actionable insights, the project enables organizations to shift from reactive problem-solving to proactive workforce management, thereby creating a more efficient and employee-centric work environment.

One of the most important impacts of this project is the improvement in **employee retention**. High attrition rates are a major concern for organizations, as they lead to increased recruitment costs, loss of skilled employees, and disruption in workflow. The analysis conducted in this project demonstrates that disengagement and burnout are key drivers of employee turnover. By identifying employees who are at risk of burnout or exhibit low engagement levels, organizations can take early interventions such as workload adjustments, counseling, or career development opportunities. This proactive approach helps in retaining valuable talent and reducing overall attrition rates.

Another critical impact is the enhancement of **employee productivity and performance**. Engaged employees are more motivated, committed, and efficient in their work. The insights derived from this project highlight the factors that influence engagement, such as job satisfaction, work-life balance, and workload. By addressing these factors, organizations can create a supportive work environment that fosters higher productivity and improved performance outcomes. This, in turn, contributes to the overall success and competitiveness of the organization.

The project also supports **data-driven decision-making in human resource management**. Traditionally, HR decisions are often based on intuition or limited information, which may not accurately reflect employee needs. The analytical framework developed in this project provides a comprehensive view of employee engagement and burnout, enabling HR managers to make informed decisions. For example, insights from the dashboard can guide decisions related to workload distribution, employee benefits, training programs, and performance management strategies.

14. CONCLUSION

This project successfully demonstrates the application of data analytics techniques in understanding and improving employee engagement, satisfaction, and burnout management within an organization. By leveraging employee-related data and transforming it into meaningful insights, the study provides a comprehensive framework for analyzing workforce behavior and identifying critical factors that influence employee well-being.

One of the key contributions of this project is the development of a unified approach to measuring employee engagement through the Engagement Index. By combining multiple satisfaction-related attributes into a single metric, the project provides a holistic view of employee experience, enabling organizations to assess engagement levels more effectively. Additionally, the introduction of the Burnout Risk Score allows for the early identification of employees who may be experiencing excessive stress due to factors such as overtime and poor work-life balance.

The findings of the analysis highlight the significant impact of workload, work-life balance, job satisfaction, and career progression on employee engagement. Employees who experience high workload and low work-life balance are more likely to exhibit lower engagement levels and higher burnout risk. These insights emphasize the importance of creating a balanced and supportive work environment to maintain employee motivation and productivity.

Another important outcome of this project is the development of an interactive dashboard using Streamlit, which provides real-time visualization of engagement metrics and burnout risk. This dashboard serves as a valuable decision-support tool for HR managers, enabling them to monitor employee well-being, identify high-risk groups, and implement targeted interventions. The ability to visualize data dynamically enhances the effectiveness of analysis and facilitates data-driven decision-making.

The project also highlights the importance of adopting a proactive approach to employee management. Traditional HR practices often focus on addressing issues after they have occurred, such as handling attrition or performance decline. In contrast, this project demonstrates how early detection of engagement issues and burnout risk can help organizations take preventive measures, thereby reducing attrition and improving overall organizational performance.

Furthermore, the integration of machine learning techniques adds predictive capabilities to the analysis, enabling organizations to anticipate potential risks and make informed decisions. Although the primary focus remains on exploratory and diagnostic analysis, the inclusion of predictive models enhances the overall value of the project.

15. REFERNCES AND PROJECT LINKS

References

The following resources were used during the development of this project:

1. Python Software Foundation. *Python Documentation*. Available at: https://docs.python.org/3/
2. Scikit-learn Developers. *Scikit-learn: Machine Learning in Python*. Available at: https://scikit-learn.org/
3. McKinney, W. (2010). *Data Structures for Statistical Computing in Python (Pandas)*.
4. Hunter, J. D. (2007). *Matplotlib: A 2D Graphics Environment*.
5. Plotly Technologies Inc. *Plotly Documentation*. Available at: https://plotly.com/
6. Streamlit Inc. *Streamlit Documentation*. Available at: https://docs.streamlit.io/
7. Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*.
8. Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques*.

Project Links

- **GitHub Repository:** https://github.com/MadhushreeSG/EduPro-Course-Demand-Prediction
- **Live Streamlit Dashboard:** https://edupro-course-demand-prediction-2tqb8spe7p4dnh5frcknyk.streamlit.app/